Seasons of Code 2026

Week 1 Submission

Task 1: Write about 5 interesting things I learnt about from the lectures.

1) I learnt that Computer vision is a field inspired by the way humans perceive and interpret visual information. By mimicking aspects of human vision, these systems can **identify patterns, detect objects, classify scenes, and make decisions based on visual data.**

2) I learnt that the human retina contains two types of photoreceptor cells: **Rods** and **Cones**. **Rods** are **highly sensitive to light** and **enable vision in dim conditions** but cannot distinguish colors. **Cones function in brighter light and are responsible for color perception**. The three cone types, sensitive to red, green, and blue wavelengths, send signals to the brain, which combines and interprets these signals to produce the full spectrum of colors we perceive.

3) I was also blown away by the fact that image resizing changes an image's dimensions by **adding, removing, or repositioning pixels.** When enlarging an image, new pixels are generated using interpolation techniques that estimate values from neighboring pixels. When shrinking, multiple pixels may be combined or discarded, causing existing pixel values to be adjusted to preserve visual quality.

4) I also learnt that computer vision uses **various color spaces** to represent image information more effectively. **RGB** represents colors through **red, green, and blue channels**, while **HSV separates hue, saturation, and brightness. Grayscale** simplifies images to intensity values, and **LAB** provides perceptually uniform color representation, improving tasks such as segmentation, object detection, and image analysis.

5) I also learnt that we can **add "noise" to a dataset to generate/ multiply its size.** For eg: We can add gaussian blur to a dataset of 1000 images, to create a set of 2000 total images, which can then be used to train a computer vision model better than the original 1000 images.

Task 2: Do a brief research on types of healthcare reports and write about medical imaging reports .

Some of the common healthcare reports include:

1. Blood Test Report:

A blood test report presents the results of laboratory analysis of blood samples. It typically includes parameters such as **red and white blood cell counts, hemoglobin levels, platelet counts, glucose levels, cholesterol levels, and electrolyte concentrations.** Physicians use these reports to assess overall health, diagnose diseases, monitor treatments, and detect abnormalities.

2. Microbiology Report:

A microbiology report **identifies microorganisms such as bacteria, viruses, fungi, or parasites** present in a patient's sample. It contains culture results, organism identification, antimicrobial susceptibility testing, and clinical interpretations. These reports help physicians diagnose infections accurately and select the most effective antibiotic or antiviral treatment.

3. Electrocardiogram (ECG/EKG) Report:

An ECG report **records and interprets the electrical activity of the heart.** It includes measurements of **heart rate, rhythm, conduction intervals, and waveform characteristics.** Cardiologists and physicians use ECG reports to detect arrhythmias, heart attacks, conduction abnormalities, and other cardiovascular disorders requiring medical attention.

4. Histopathology Report:

A histopathology report is a **specialized pathology report focused on microscopic examination of tissue samples**. It describes tissue architecture, cellular abnormalities, disease progression, and malignancy characteristics. Histopathology reports play a central role in cancer diagnosis, grading, staging, and treatment planning by providing detailed insights into tissue-level changes.

5. Pathology Report:

A pathology report provides the results of laboratory **examination of tissues, cells, or bodily fluids.** Commonly generated after biopsies or surgical procedures, it includes specimen descriptions, microscopic findings, diagnoses, and staging information when applicable. Pathology reports are crucial for **diagnosing cancers, infections, inflammatory diseases, and various other medical conditions**.

More on medical imaging reports:

Medical imaging plays a fundamental role in modern healthcare by enabling clinicians to **visualize internal organs, tissues, and physiological processes** without invasive procedures. Techniques such as **X-ray imaging, CT, MRI, ultrasound, and PET scans** generate vast amounts of diagnostic data that support disease detection, treatment planning, and patient monitoring. Traditionally, radiologists analyze these images and document their findings in detailed medical imaging reports, which serve as a critical communication link between imaging specialists and treating physicians. In recent years, advances in computer vision and deep learning, particularly through CNNs and Vision Transformers, have enabled **automated image analysis and report-generation systems** that can assist radiologists in handling the growing volume of medical imaging data.